MUHAMMED ANWAF

Backend Development | AWS | Applied AI

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Portfolio | GitHub: muhd-anwaf | LinkedIn | LeetCode

Summary

NIT Calicut Mechanical Engineering graduate with project experience building Java/Spring Boot APIs, PostgreSQL-backed applications, and Python AI services. Deployed an e-commerce backend on AWS Elastic Beanstalk with Amazon RDS; built payment verification workflows, a document RAG assistant, and a containerized ML inference API with automated CI tests.

Technical Skills

Languages: Java, Python, SQL, JavaScript, TypeScript
Backend: Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST APIs, JWT, FastAPI
Databases & Cloud: PostgreSQL, Supabase, AWS Elastic Beanstalk, Amazon RDS, Docker, Vercel
AI & ML: LangChain, FAISS, Gemini API, RAG, Pydantic, scikit-learn, MLflow, Pandas, NumPy
Tools & Foundations: Git, GitHub Actions, Maven, pytest, Data Structures & Algorithms, OOP
Frontend: React, Next.js, Tailwind CSS

Projects

E-Commerce Backend Service GitHub

Java, Spring Boot, Spring Security, PostgreSQL, AWS

- Built REST APIs for products, categories, carts, orders, addresses, and authentication using layered controllers, services, and repositories with Spring Data JPA.
- Added product search, pagination, sorting, request validation, and centralized exception handling; implemented stateless JWT authentication and BCrypt password hashing.
- Deployed the backend on AWS Elastic Beanstalk with Amazon RDS for PostgreSQL persistence.

Berry Organics – E-Commerce Platform Demo | GitHub

Next.js, TypeScript, Supabase, PostgreSQL, Razorpay, Vercel

- Built guest/customer checkout and protected admin workflows; recalculated prices and validated stock server-side using current database records instead of browser-supplied totals.
- Integrated Razorpay Test Mode with server-created payment orders, HMAC-SHA256 signature verification, and a PostgreSQL RPC for payment finalization and inventory updates.
- Implemented customer order history, admin product/order management, and Supabase image uploads with cleanup of replaced or unused files.

RAG-Based Document Q&A Assistant Demo | GitHub

Python, LangChain, Gemini API, FAISS, Streamlit, Pydantic

- Built PDF ingestion, overlapping text chunking, and FAISS retrieval with 768-dimensional embeddings to generate answers with source/page citations.
- Batched embedding requests in groups of up to 100 chunks and reused the document index in Streamlit session state across questions.
- Validated structured responses and citation IDs; added unsupported-answer fallback and four test cases checking grounding and instruction-override behavior.

Walmart Sales Forecast – ML Inference & CI GitHub

Python, FastAPI, Pydantic, scikit-learn, Docker, GitHub Actions

- Built a FastAPI prediction endpoint with Pydantic input validation and model loading at application startup; containerized the inference service.
- Configured GitHub Actions to run pytest on pushes and pull requests, with tests for API, data processing, and model training.

Education

National Institute of Technology Calicut 2022–2026
B.Tech in Mechanical Engineering | CGPA: 6.98/10
Relevant coursework: Machine Learning for Data Science (S Grade), Neural Networks for CFD

Professional Development

AWS Cloud Practitioner Essentials – AWS Skill Builder, 2026
Deep Learning with PyTorch, Keras and TensorFlow – IBM/Coursera, 2026